INCH-POUND

MIL-R-39007G
16 May 1991
SUPERSEDING
MIL-R-39007F
15 JANUARY 87

MILITARY SPECIFICATION
RESISTORS, FIXED, WIRE-WOUND (POWER TYPE), ESTABLISHED RELIABILITY GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for axial lead precision, power type, wire-wound, fixed resistors having a +25°C ambient operating temperature derated to zero load at +275°C for use in electrical, electronic communication, and associated equipment. These resistors have initial resistance tolerances of ±.5, ±.1, and ±1 percent, and a resistance temperature characteristic as given in table VII. Resistors covered by this specification shall have failure rates ranging from 1 to .001 percent per 1,000 hours (see 1.2.1.5). These failure rates are established at 60 percent confidence on the basis of life tests (permissible resistance change of 1.0 percent). The failure rate, identified by the appropriate symbol (see 1.2.1.5) refers to operation at full rated wattage (or voltage) at +25°C ambient temperature (see 6.1). Table I of supplement 1 provides a summary of performance characteristics for these resistors. Resistors which are noninductively wound with solderable terminals are designated by a terminal designated "N" and "Z". Terminals designated by the letters "S" and "W" are inductively wound with solderable and weldable terminals (see 1.2.1.2). A part per million (PPM) quality system is used for documenting and reporting the average outgoing quality of resistors supplied to this specification. Statistical process control (SPC) techniques are required in the manufacturing process to minimize variation in production of resistors supplied to the requirements of this specification.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). The PIN shall be in the following format, and as specified (see 3.1 and 6.1):

RWR74	S	49R9	F	M
Style (See 1.2.1.1)	Terminal wire and winding (See 1.2.1.2)	Resistance (See 1.2.1.3)	Resistance tolerance (See 1.2.1.4)	Failure rate (See 1.2.1.5)

1.2.1.1 Style. The style shall be identified by the three-letter symbol "RWR" followed by a two digit number; the letters identify established-reliability, axial-lead, power type, wire-wound, fixed resistors, and the number identifies the size, wattage rating, and construction of the resistors.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5905

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

1.2.1.2 Terminal and winding. The terminal capability and inductive or noninductive winding designation are identified by a single letter in accordance with table I.

TABLE I. Terminal and winding.

Symbol	Terminal and winding
S	Solderable, inductively wound
W	Weldable, inductively wound
N	Solderable, noninductively wound 1/
Z	Weldable, noninductively wound 1/

1/ Wound using Ayrton-Perry method.

1.2.1.3 Resistance. The nominal resistance is expressed in ohms and is identified by four digits; the first three digits represent significant figures and the last digit specifies the number of zeros to follow. When the value of resistance is less than 100 ohms, or when fractional values of an ohm are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, succeeding digits of the group represent significant figures. The resistance value designations are shown in table II. Minimum and maximum resistance values shall be as specified (see 3.1). The standard resistance values for every decade shall follow the sequence demonstrated for 10 to 100 decade as specified in table III. For resistance tolerances F and D, only those resistance values which follow the sequence of values listed in the 10 to 100 decade specified in table III shall be considered as conforming to this specification.

TABLE II. Designation of resistance values.

Designation	Resistance (ohms)
1R00 to 9R88 inclusive	1 to 9.88 inclusive
10R0 to 98R8 inclusive	10 to 98.8 inclusive
1000 to 9880 inclusive	100 to 988 inclusive
1001 to 9881 inclusive	1,000 to 9,880 inclusive
1002 to 4022 inclusive	10,000 to 40,200 inclusive

Source: https://assist.dla.mil -- Downloaded: 202610-21T16:22Z
Check the source to verify that this is the currentversion before use.

TABLE III. Standard resistance values for the 10 to 100 decade.

0.1 (B) 1/	0.5 (D)	1.0 (F)	0.1 (B) 1/	0.5 (D)	1.0 (F)	0.1 (B) 1/	0.5 (D)	1.0 (F)
10.00	10.00		22.90	---		49.30	---	
10.10	---		23.20	23.20		49.90	49.90	
10.20	10.20		23.40	---		50.50	---	
10.40	---		23.70	23.70		---	---	
10.50	10.50		24.00	---		51.10	51.10	
10.60	---		24.30	24.30		51.70	---	
10.70	10.70		24.60	---		52.30	52.30	
10.90	---		24.90	24.90		53.00	---	
11.00	11.00		25.20	---		53.60	53.60	
11.10	---		25.50	25.50		54.20	---	
11.30	11.30		25.80	---		54.90	54.90	
11.40	---		26.10	26.10		55.60	---	
11.50	11.50		26.40	---		---	---	
11.70	---		26.70	26.70		56.20	56.20	
11.80	11.80		---	---		56.90	---	
12.00	---		27.10	---		57.60	57.60	
12.10	12.10		27.40	27.40		58.30	---	
12.30	---		27.70	---		59.00	59.00	
12.40	12.40		28.00	28.00		59.70	---	
12.60	---		28.40	---		60.40	60.40	
12.70	12.70		28.70	28.70		61.20	---	
12.90	---		29.10	---		61.90	61.90	
13.00	13.00		29.40	29.40		---	---	
13.20	---		29.80	---		62.60	---	
13.30	13.30		---	---		63.40	63.40	
13.50	---		30.10	30.10		64.20	---	
13.70	13.70		30.50	---		64.90	64.90	
13.80	---		30.90	30.90		65.70	---	
14.00	14.00		31.20	---		66.50	66.50	
14.20	---		31.60	31.60		67.30	---	
14.30	14.30		32.00	---		---	---	
14.50	---		32.40	32.40		68.10	68.10	
14.70	14.70		32.80	---		69.00	---	
14.90	---		---	---		69.80	69.80	
15.00	15.00		33.20	33.20		70.60	---	
15.20	---		33.60	---		71.50	71.50	
15.40	15.40		34.00	34.00		72.30	---	
15.60	---		34.40	---		73.20	73.20	
15.80	15.80		34.80	34.80		74.10	---	
16.00	---		35.20	---		75.00	75.00	
16.20	16.20		35.70	35.70		75.90	---	
16.40	---		---	---		76.80	76.80	
16.50	16.50		36.10	---		77.70	---	
16.70	---		36.50	36.50		78.70	78.70	
16.90	16.90		37.00	---		79.60	---	
17.20	---		37.40	37.40		80.60	80.60	
17.40	17.40		37.90	---		81.60	---	
17.60	---		38.30	38.30		---	---	
17.80	17.80		38.80	---		82.50	82.50	

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

TABLE III. Standard resistance values for the 10 to 100 decade - Continued.

0.1 (B) 1/	0.5 (D)	1.0 (F)	0.1 (B) 1/	0.5 (D)	1.0 (F)	0.1 (B) 1/	0.5 (D)	1.0 (F)
18.00	18.00		---	---		83.50	---	
18.20	18.20		39.20	39.20		84.50	84.50	
18.40	---		39.70	---		85.60	---	
18.70	18.70		40.20	40.20		86.60	86.60	
18.90	---		40.70	---		87.60	---	
19.10	19.10		41.20	41.20		88.70	88.70	
19.30	---		41.70	---		89.80	---	
19.60	19.60		42.20	42.20		90.90	90.90	
19.80	---		42.70	---		---	---	
20.00	20.00		---	---		92.00	---	
20.30	---		43.20	43.20		93.10	93.10	
20.50	20.50		43.70	---		94.20	---	
20.80	---		44.20	44.20		95.30	95.30	
21.00	21.00		44.80	---		96.50	---	
21.30	---		45.30	45.30		97.60	97.60	
21.50	21.50		45.90	---		98.80	---	
21.80	---		46.40	46.40				
---	---		47.00	---				
22.10	22.10		47.50	47.50				
22.30	---		48.10	---				
22.60	22.60		48.70	48.70				

1/ The resistance values for tolerance B may be of any value, but it is preferred that the values be chosen from the tolerance D values in this specification.

1.2.1.4 Resistance tolerance. The resistance tolerance is identified by a single letter in accordance with table IV.

TABLE IV. Resistance tolerance.

Symbol	Resistance tolerance
	Percent (±)
B	0.1
D	0.5
F	1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

1.2.1.4.1 Replacement or maintenance (applicable to 0.1-and 0.5-percent tolerances). It is preferred that resistance values be selected as specified in table III. However, if it is impracticable to do so, then any value (within specification limits) may be specified and these values shall be considered as conforming to this specification (see table V and 3.1).

TABLE V. Designation of resistance values (0.1- and 0.5-percent tolerances).

Designation	Resistance value
R100 to R999 inclusive	.1 to .9 inclusive
1R00 to 9R99 inclusive	1 to 9 inclusive
10R0 to 99R9 inclusive	10 to 99 inclusive
1000 to 9990 inclusive	100 to 999 inclusive
1001 to 9991 inclusive	1,000 to 9,999 inclusive
1002 to maximum	10,000 to maximum

1.2.1.5 Failure rate. The failure rate is identified by a single letter in accordance with table VI.

TABLE VI. Failure rate (60-percent confidence).

Failure-rate-level designation	Failure rate percent/1,000 hours
M	1.0
P	0.1
R	0.01
S	0.001

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-39032 - Resistors, Packaging of.

(See supplement 1 for a list of associated specifications.)

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Reliability Assurance Program for Electronic Parts Specifications.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-1276 - Leads for Electronic Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-45662 - Calibration Systems Requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-21T16:22Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Non-Government publications. The following documents, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION

EIA-554 - Assessment of Outgoing Nonconforming Levels in Part Per Million (PPM) (DoD adopted).
EIA-557 - Statistical Process Control Systems (DoD adopted).

(Application for copies should be addressed to Electronic Industries Association, 2001 Eye Street, Washington D.C. 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.2 Order of precedence. In the event of a conflict between the text of this document and references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Detail requirements. Detail requirements or exceptions applicable to individual types of resistors shall be as specified in the detail specifications listed in supplement 1 to this specification. In the event of any conflict between requirements of this specification and detail specifications, the latter shall govern (see 6.1).

3.2 Qualification. Resistors furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List (QPL) at the time set for opening of bids (see 4.4 and 6.2). In addition, the manufacturer shall obtain certification from the qualifying activity that the reliability assurance requirements of 4.1.3 have been met and are being maintained. Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL.

3.3 Reliability and quality.

3.3.1 Reliability. Reliability of resistors furnished under this specification shall be established and maintained in accordance with procedures and requirements specified in MIL-STD-790 and MIL-STD-690 with details and exceptions specified in 4.1.3, 4.4.4.2, 4.5, and 4.6.1.3.

3.3.2 Quality.

3.3.2.1 Statistical process control (SPC). The contractor shall implement and use statistical process control techniques in the manufacturing process for parts covered by this specification. The SPC program shall be developed and maintained in accordance with EIA-557. The SPC program shall be documented and maintained as part of the overall reliability assurance program as specified in MIL-STD-790. The implementation date for statistical process control shall be 12 months from the date of this specification. Processes for application of SPC techniques should include but are not limited to:

a. Monitoring dc resistance values.
b. Lead attachment "axial pull test".
c. Monitoring dc resistance values.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

3.3.2.2 Quality levels. The quality of lots that have been subject to and passed the subgroup 1 100-percent screening inspection of the group A inspection shall be established and maintained in accordance with 4.6.1.2.3 and EIA-554. Individual PPM defect levels (i.e., PPM-2 and PPM-3) and an overall PPM defect level (i.e., PPM-5) shall be established based on the tests prescribed in the subgroup 2 tests of the group A inspections. The defect level for PPM-2 shall be less than 100 PPM. Data shall not be excluded from the appropriate PPM calculation unless specifically authorized by the qualifying activity. Guidance for exclusion of data is specified in EIA-554. The implementation date for PPM verification shall be 12 months from the date of this specification.

3.3.2.2.1 Noncompliance. The contractor shall notify the qualifying activity when the 100 PPM level is reached or exceeded for PPM-2. The contractor shall provide sufficient information to the qualifying activity documenting the causes of the problem and what corrective action is being taken. Failure to correct this problem shall be the basis for removal of the affected product from the QPL.

3.4 Material. The material shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the resistors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.5 Design and construction. Resistors shall be of the design, construction, and physical dimensions specified (see 3.1). The design of the resistors shall be such as to preclude shorting of turns and to obtain a minimum voltage drop between adjacent turns. Except as required for noninductive performance capability, resistors shall be wound with a single layer of resistance wire. For terminal wire and winding symbols "N" and "Z" resistors shall be wound by the Ayrton-Perry method.

3.5.1 Windings.

3.5.1.1 Wire. The manufacturer shall verify by certification that the wire has no joints, welds, or bonds, except at the end terminals where welding, brazing, or silver solder only shall be employed. The wire shall possess a uniform cross section of conductor and insulation (if employed), and shall be as free as practicable from particles or impurities, grain growth, or other factors contributing to stop weakness. The cross sectional area of the wire shall be the maximum consistent with other requirements specified herein. Where abrasive resistance value adjustment is used, the abrasion must be evenly distributed around the resistor body. The remaining cross sectional area of the wire, after the abraiding operation, shall be at least 90 percent of the original cross sectional area.

3.5.1.2 Pitch. The average winding pitch shall not exceed five times the wire diameter.

3.5.1.3 Effective wire coverage. Effective wire coverage is the winding length on the tube between points of departure from the nominal winding pitch. The effective wire coverage shall be such that not more than 20 percent of the overall winding area shall remain uncovered (by wire).

3.5.1.4 Protective coating or enclosure. Resistor assemblies shall be protected by a coating of moisture resistant insulating material which shall completely cover the exterior of the resistance element, including connections or terminations. The coating shall not crack, craze, run, or form globules at any temperature up to and including +275°C, regardless of the mounting position of the resistor. This material shall afford adequate protection against the effects of prolonged exposure to high humidities. The protective coating or enclosure shall be such as to minimize the establishment of leakage paths between the terminals resulting from collection of moisture film on the exterior surface of the resistor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

3.5.1.5 Terminals. Terminals shall be made of a solid conductor of the length and diameter specified.

3.5.1.5.1 Solderable terminals. Solderable leads shall be suitably treated to meet the requirements of solderability (see 4.7.4). At the option of the manufacturer, the terminals may be solder coated or otherwise treated to meet solderability requirements following the conditioning test specified in 4.7.2. When a lead coating containing tin is used, the tin content shall range between 40 and 70 percent (see 3.1).

3.5.1.5.2 Weldable terminals. Weldable terminals shall be type N-1 of MIL-STD-1276. The manufacturer shall verify by certification that the weldable terminals meet all the requirements for type N-1 of MIL-STD-1276 (see 3.1). The solderability requirement of 3.10 is not applicable to weldable terminals.

3.5.1.6 Solder dip (retinning) leads. The manufacturer (or his authorized category B or C distributor) may solder dip/retin the leads of product supplied to this specification, provided the solder dip process has been approved by the qualifying activity.

3.5.1.6.1 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original lead finish qualified was hot solder dip lead finish 52 of MIL-STD-1276 (NOTE: The 200 microinch maximum thickness is not applicable), the manufacturer shall use the same solder dip process for retinning as used in the original manufacture of the product.

b. When the lead originally qualified was not hot solder dip lead finish 52 of MIL-STD-1276 as prescribed in a., approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder dip process, the resistors are subjected to the dc resistance test. No defects are allowed.

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to solder heat test followed by the moisture resistance test (NOTE: Solder dip of gold plated leads is not allowed).

3.5.1.6.2 Solder dip/retinning options. The manufacturer (or authorized category B or C distributor) may solder dip/retin as follows:

a. After the 100-percent group A screening tests. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100-percent screening tests shall be repeated on 100 percent of the lot (NOTE: The manufacturer may solder dip/retin prior to the 100-percent electrical measurements of the group A, subgroup 1 tests). The Percentage Defective Allowable (PDA) for the electrical measurements shall be as for the subgroup 1 tests.

b. As a corrective action if the lot fails the group A solderability test.

c. After the group A inspection has been completed. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100-percent screening test shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as specified for the subgroup 1 tests. Following these tests, the manufacturer shall submit the lot to the group A solderability test as specified in 4.7.4.

Source: https://assist.dla.mil -- Downloaded: 202610-21T16:22Z
Check the source to verify that this is the current version before use.

3.5.1.7 Weight. Resistors shall not exceed the maximum weight specified (see 3.1).

3.6 Power rating. Resistors shall have a power rating based on continuous full load operation at an ambient temperature of +25°C (see 3.1). This power rating is dependent on the ability of resistors to meet the failure rate requirements specified in 3.22. For temperatures in excess of those specified above, the load shall be derated in accordance with figure 6.

3.7 Voltage rating. Resistors shall have a rated direct current (dc) continuous working voltage, or an approximate sine wave root mean square (rms) continuous working voltage at commercial line frequency and waveform corresponding to the power rating, as determined from the following formula:

$$E = \sqrt{PR}$$

Where: E = rated dc or rms continuous working voltage
P = power rating (see 3.1)
R = nominal resistance

3.8 Conditioning. When resistors are tested as specified in 4.7.2, there shall be no mechanical damage. The change in resistance shall not exceed ±(0.2 percent +.005 ohm).

3.9 DC resistance. When resistors are tested as specified in 4.7.3, the dc resistance shall be within the specified tolerance of the nominal resistance (see 1.1 and 3.1).

3.10 Solderability (applicable to S and M terminals only). When resistors are tested as specified in 4.7.4, they shall meet the criteria for wire lead terminal evaluation in the test method.

3.11 Resistance to solvents. When resistors are tested as specified in 4.7.5, there shall be no evidence of mechanical damage and marking shall remain legible.

3.12 Thermal shock. When resistors are tested as specified in 4.7.6, there shall be no evidence of mechanical damage; the change in resistance shall not exceed ±(0.2 percent +.005 ohm).

3.13 Resistance-temperature characteristic. When resistors are tested as specified in 4.7.7, the resistance temperature characteristic between -55°C and +275°C shall not exceed values as shown in table VII.

TABLE VII. Resistance temperature characteristic.

Resistance range	Resistance temperature characteristic parts per million/°C
0.1 to 0.499 inclusive	±650
0.499 to 1.0 inclusive	±400
1.0 to 10 inclusive	±50
10 and above inclusive	±20

3.14 Low-temperature storage. When resistors are tested as specified in 4.7.8, there shall be no evidence of mechanical damage. The change in resistance shall not exceed ±(0.1 percent +.005 ohm).

3.15 Short-time overload. When resistors are tested as specified in 4.7.9, there shall be no evidence of arcing, burning, or charring; the change in resistance shall not exceed ±(0.2 percent +.005 ohm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

3.16 Dielectric withstanding voltage. When resistors are tested as specified in 4.7.10, the leakage rate shall not exceed 500 microamperes; there shall be no evidence of flashover, mechanical damage, arcing or insulation breakdown; the change in resistance shall not exceed ±(0.1 percent +.005 ohm).

3.17 Insulation resistance. When resistors are tested as specified in 4.7.11, the insulation resistance shall not be less than 1,000 megohms.

3.18 Moisture resistance. When resistors are tested as specified in 4.7.12, there shall be no evidence of mechanical damage; the change in resistance shall not exceed ±(0.2 percent +.005 ohm). In addition, the dielectric withstanding voltage shall be as specified in 3.16, and the insulation resistance shall be 100 megohms, minimum.

3.19 Terminal strength. When resistors are tested as specified in 4.7.13, there shall be no evidence of breaking or loosening of terminals from the resistor form, or chipping of the coating on the leads is permissible as long as the end caps are not exposed. The change in resistance shall not exceed ±(0.1 percent +.005 ohm).

3.20 Shock (specified pulse). When resistors are tested as specified in 4.7.14, there shall be no evidence of mechanical or electrical damage; the change in resistance shall not exceed ±(0.1 percent +.005 ohm). There shall be no electrical discontinuity during the tests.

3.21 Vibration, high frequency. When resistors are tested as specified in 4.7.15, there shall be no evidence of mechanical damage; the change in resistance shall not exceed ±(0.1 percent +.005 ohm). There shall be no electrical discontinuity during the test.

3.22 Life. When resistors are tested as specified in 4.7.16, the change in resistance between the initial measurement and by any succeeding measurement shall not exceed the specified value for the application test period; there shall be no evidence of mechanical damage.

3.22.1 Qualification inspection. When resistors are tested as specified in 4.7.16, the change in resistance between the initial measurement and any succeeding measurement up to and including 2,000 +72, -0 hours, shall not exceed ±0.5 percent. The entire qualification sample shall be continued on test for 10,000 +96, -0 hours.

3.22.2 Failure rate determination (extended FR test). When resistors are tested as specified in 4.7.16, the change in resistance between the initial measurement up to and including 10,000 +96, -0 hours shall not exceed ±1.0 percent. This single failure criteria shall be applicable to all measurements during the life test for purposes of determining failure rate level qualification and is applicable as a parallel requirement with 3.22.1 to the measurement made during the life tests specified for qualification inspection.

3.23 High temperature exposure. When resistors are tested as specified in 4.7.17, there shall be no evidence of mechanical damage and the change in resistance shall not exceed ±0.5 percent for values above 100 ohms and ±(0.5 percent +.05 ohm) for values 100 ohms and below.

3.24 Fungus. All external materials shall be nonnutrient to fungus growth or shall be suitably treated to retard fungus growth. The manufacturer shall verify by certification that all external materials are fungus resistant or shall test the resistors as specified in 4.7.18. There shall be no evidence of fungus growth on the external surfaces.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

3.25 Reactance (applicable to "N" and "Z" terminals and windings only). When resistors are tested as specified in 4.7.19, they shall be within the maximum limits specified in table VIII.

TABLE VIII. Reactance.

Styles	Maximum effective series inductance uH		Maximum effective parallel capacitance - pF
	Below 50 ohms	Above 50 ohms	All resistance values
RWR71	0.20	0.37	1.5
RWR74	0.35	0.60	1.5
RWR78	0.65	1.20	1.5
RWR80	0.20	0.37	1.5
RWR81	0.20	0.37	1.5
RWR82	0.20	0.37	1.5
RWR84	0.30	0.60	1.5
RWR89	0.20	0.37	1.5

3.26 Marking. Resistors shall be marked with the PIN, "JAN" marking, date code, source code, and manufacturer's production lot code (see MIL-STD-790). Date and source code shall be in accordance with MIL-STD-1285. At the option of the manufacturer, the PIN shall be divided between the characteristic letter and the first digit of the resistance value. The following is an example of complete marking:

12345 - Source code
6633J - Date code, lot code (any letter except "J"), and "JAN" marking
RWR74 - Style.
N1000 - Terminal-noninductive designator and resistance type designation.
FM - Tolerance and failure rate.

The date code shall provide traceability through all production operations and shall represent the date of winding or enclosure as consistently provided by the manufacturer.

3.26.1 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over, the certification marks "JAN and J", respectively, to indicate electrical equipment, namely resistors, capacitors, electron tubes, and the like, acquired by, or manufactured for use by, of for the Government in accordance with standard government specifications. Accordingly, resistors acquired to, and meeting all of the criteria specified herein and in applicable detail specification shall bear the certification mark "JAN", except that resistors too small to bear the certification mark "JAN" shall bear the letter "J". The "JAN" or "J" shall be placed immediately before the type number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN" or"J" may be located on the first line above or below the PIN. Resistors furnished under contract or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable detail specifications shall not bear "JAN" or "J". In the event a resistor sample fails to meet the requirements of this specification and applicable detail specification, the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all resistors represented by sample. The United States Government has obtained certificate of registration number 504,860 for the certification for the certification mark "JAN" or "J".

3.26.2 Minimum marking. When the physical size of the resistor style precludes the marking of all of the information cited in the example in 3.26, the minimum marking required shall be as specified in the detail specification (see 3.1). Marking shall remain legible at the end of all tests. In those cases where full marking requirements are not on the resistor body, full marking shall be marked on the unit package.

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T16:22Z
Check the source to verify that this is the currentversion before use.

3.26.3 Use of conductive inks. Conductive inks shall not be used to coat the body of the resistor as preparation for marking or used for marking of resistors.

3.27 Supplying to higher failure rate levels. A manufacturer may supply to all higher failure rate levels than to which he is qualified. Parts qualified and marked to lower failure rate levels are substitutable, with acquiring agency approval, for higher failure rate level parts, and shall not be remarked unless specified in the contract or purchase order (see table IX)(see 6.1).

TABLE IX. Failure rate substitution.

Failure rate	Failure rate substitute
S (.001)	---
R (.01)	S
P (0.1)	S, R,
M (1.0)	S, R, P

3.27.1 Supplying to higher resistance tolerances. Parts qualified and marked to lower tolerances, with acquiring agency approval, are substitutiable for parts marked to higher tolerance levels and shall not be remarked unless specified in the contract or purchase order (see 6.1) (see table X).

TABLE X. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitution
B	---
D	B
F	B, D

3.28 Workmanship. Resistors shall be processed in such a manner as to be uniform in quality and shall meet the requirements of 3.4 to 3.5.1.7, inclusive, and shall be free from defects that will affect, life, serviceability, or appearance.

3.28.1 Soldering. Where soldering is employed, only noncorrosive fluxes shall be used, unless it can be shown that corrosive elements have been satisfactorily removed after soldering. Solder shall not be used for obtaining mechanical strength. Electrical connections shall be mechanically secure before soldering and electrically continuous after soldering. Except for solder used to coat the terminals, the solder used shall in no case start to melt at a temperature less than +300°C.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to, which permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-STD-45662.

4.1.3 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance with MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.1.4 Statistical process control. An SPC program shall be established and maintained in accordance with EIA-557. Evidence of such compliance shall be verified by the qualifying activity as a prerequisite for qualification and retention of qualification.

4.2 Classification of inspection. The examination and testing of resistors shall be classified as follows:

a. Qualification inspection (see 4.4).

b. Verification of qualification (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions and precautions.

4.3.1 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.2 Precautions. Adequate precautions shall be taken during inspection to prevent condensation of moisture on resistors, except during the moisture-resistance test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.2).

TABLE XI. Qualification inspection.

Inspection	Requirements paragraph	Method paragraph	Number of samples 1/	Number of defects permitted 2/	
Group I					
Conditioning 3/	3.8	4.7.2			
DC resistance 3/	3.9	4.7.3		4/	
Group IA (coated or enclosed)	3.1, 3.4 to 3.5.1.6 inclusive		296		
Visual and mechanical inspection 5/	3.26 to 3.28 inclusive	4.7.1		1	
Group II					
Solderability (where applicable)(both leads)	3.10	4.7.4	10 6/	1	
Resistance to solvents	3.11	4.7.5			
Group III					
Thermal shock 7/	3.12	4.7.6			
Resistance temperature characteristic 7/	3.13	4.7.7	32 (16 highest, 16 lowest)		
Low temperature storage	3.14	4.7.8			
Short time overload 7/	3.15	4.7.9		1	1
Dielectric withstanding voltage 7/	3.16	4.7.10			
Insulation resistance	3.17	4.7.11			
Moisture resistance	3.18	4.7.12			
Terminal strength	3.19	4.7.13			
Group IV					
Thermal shock	3.12	4.7.6	30 (15 highest, 15 lowest)		
Shock (specified pulse)	3.20	4.7.14		1	
Vibration, high frequency	3.21	4.7.15			
Group V (uncoated or unenclosed) 8/					
Visual and mechanical examination	3.1, 3.4, 3.5, 3.26, 3.26.1	4.7.1, as applicable	4 (2 highest, 2 lowest)	0	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

TABLE XI. Qualification inspection - Continued.

Inspection	Requirements paragraph	Method paragraph	Number of samples 1/	Number of defects permitted 2/
Group VI				
Life	3.22	4.7.16	102 (51 highest, 51 lowest)	1
Group VII				
High temperature exposure	3.23	4.7.17	102 (51 highest, 51 lowest)	1
Group VIII				
Fungus	3.24	4.7.18	10	0
Group IX				
Reactance	3.25	4.7.19	30 (10 lowest, 10- 301 ohms, 10 highest)	0

1/ See appendix for details.
2/ Failure of a resistor in one or more tests of a test group shall be charged as a single failure.
3/ Tests shall not be performed if a manufacturer presents certified data proving tests have been performed on the qualification sample.
4/ Items subjected to groups IA through IX inclusive, must meet requirements of group I.
5/ Marking shall be considered defective only if illegible or missing. Marking shall remain legible at the end of all tests.
6/ Any nine samples from group II shall be subjected to resistance to solvents.
7/ Nondestructive tests.
8/ Unenclosed sample units shall be subjected to visual and mechanical inspection as specified in group IA only.

4.4.1 Sample. The sample shall be taken from a production run and shall be produced with equipment and procedures normally used in production. Each resistor style shall be qualified separately.

4.4.2 Test routine. Sample units shall be subjected to the qualification inspection specified in table XI in the order shown. All sample units with the exception of those for groups II and VIII inspections, shall be subjected to the inspections of group IA. The 296-sample units shall then be divided as specified in table XI for groups III, IV, VI, VII, and IX and subjected to the inspections as applicable. In addition, four uncoated or unenclosed sample units shall be selected and subjected to group V inspection. An additional 10 sample units each shall be subjected to the first inspection of group II and nine of these same units shall be subjected to the second test of group II. An additional 10 samples shall be subjected to the tests of group VIII. Additional samples for tolerance qualification shall be as specified in 30.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.4.3 Defectives. Defectives in excess of those allowed in table XI shall be cause for refusal to grant qualification.

4.4.4 Failure rate level and quality level verification.

4.4.4.1 FR qualification. FR qualification shall be in accordance with the general and detailed requirements of MIL-STD-690 and the following details:

a. Procedure I - Qualification at the initial FR level. Level M (1.0 percent) of FRSP-60 shall apply. Sample units shall be subjected to the qualification inspection specified in group VI, table XI (see 4.4.2). Entire life test sample shall be continued on test to 10,000 hours as specified in 4.7.16, upon completion of the 2,000-hour qualification.

b. Procedure II - Extension of qualification to lower FR levels. To extend qualification to the R (0.01 percent) and S (0.001 percent) FR levels, data from two or more styles of similar construction may be combined. Style combinations shall be as described for lot formation (see 4.6.1.1).

c. Procedure III - Maintenance of FR levels qualification. Maintenance period A of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification (see 4.6.2 group C inspection).

4.4.4.2 Quality level verification. The contractor is responsible for establishing a quality system to verify the PPM defect level of lots that are subjected to subgroup 2 tests of the group A inspections. The PPM defect level shall be based on a 6 month moving average. The contractor shall verify and report monthly the individual PPM categories (i.e., PPM-2 and PPM-3) and an overall PPM defect level (i.e., PPM-5).

4.5 Verification of qualification. Every 6 months the manufacturer shall compile a summary of the results of quality conformance inspections and extended FR test data, in the form of a verification of qualification report, and forward it to the qualifying activity within 30 days after the end of the reporting period, as the basis of continued qualification approval. In addition, the manufacturer shall immediately notify the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain the qualified FR level or the group C inspection data indicates failure of the qualified product to meet the requirements of this specification. Continuation shall be based on evidence that over a 6-month period the following conditions have been met:

a. Verification by the qualifying activity that the manufacturer meets the requirements of MIL-STD-790.

b. The manufacturer has not modified the design of the item.

c. The specification requirements for the item have not been amended so far as to affect the character of the item.

d. Lot rejection for group A, subgroup 1 inspection does not exceed 5 percent or one lot, whichever is greater (subgroups 2, 3, and 4 of group A are not included in the reject rate).

e. Lot rejection for group B inspection does not exceed 5 percent or one lot, whichever is greater.

f. The requirements for group C inspection are met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

g. The records of FR tests combined substantiate that the "M" (1.0 percent), or "P" (0.1 percent) FR level has been maintained, or that the manufacturer continues to meet the "R" (0.01 percent), and "S" (0.001 percent) FR levels for which qualified, although the total component hours of testing do not, as yet, meet the requirement of 4.4.4.1c.

h. The contractor shall provide documentation to the qualifying activity pertaining to PPM calculations including numbers of part types tested, individual PPM defect categories (i.e., PPM-2 and PPM-3) and the overall PPM defect rate (PPM-5). This information shall be submitted on a detail specification basis and based on a 6-month moving average.

When group C requirements are not met and the manufacturer has taken corrective action satisfactory to the Government, group C retesting shall be instituted. A summary of retesting shall be forwarded to the qualifying activity within 30 days after completion of the retest. All reports are to be certified by a responsible company official and Government inspector.

If group C test requires a "post-test" readings with initial readings (delta measurements), the verification of qualification summary shall include the most positive and most negative delta change for each inspection lot. For life testing, delta R readings shall be reported at each interval in which readings are taken.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspections. When specified, a copy of the test data, for groups A and B shall be certified by a responsible company official of the manufacturer and forwarded with each lot shipped.

4.6.1.1 Inspection and production lot.

4.6.1.1.1 Inspection lot. The total number of pieces in any given lot submitted for quality conformance inspection shall not exceed the number of pieces accumulated from one month's production. Resistors of different styles, resistance values, and tolerances may be combined to form a lot provided they are all made with the same enclosure material on the same production line using the same methods and processes. Resistors with different core, cap, and terminal materials that have been qualified, and different wire materials may be combined in the same lot. Noninductively wound resistors designated "N" may be combined with resistors which are not noninductively wound (see 3.5 and 1.2.1.2).

4.6.1.1.2 Production lot. A production lot consists of parts manufactured from the same basic raw materials, processed under the same specifications and procedures, and produced with the same equipment. Each production lot of parts should be a group identified by a common manufacturing record through all significant manufacturing operations.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspection specified in table XII, and shall be made on the same set of sample units in the order shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.6.1.3 Quality supply records shall be appropriately annotated

TABLE XII. Group A inspection.

Inspection	Requirement paragraph	Method paragraph	Sampling procedure
Subgroup 1 1/			
Conditioning	3.8	4.7.2	100-percent inspection
DC resistance 2/	3.9	4.7.3	
Subgroup 2 (PPM)			
DC resistance (PPM-2)	3.9	4.7.4	See table XIII
Mechanical (PPM-3) (Dimensions only)	3.1	4.7.1	
Subgroup 3			
Visual examination	3.1, 3.4, 3.5.1.7 and 3.26 incl, and 3.28 to 3.28.1,	4.7.1	See 4.6.1.2.4
Terminals	3.5.1.5 to 3.5.1.6.2, incl.	4.7.1	
Marking	3.26, 3.26.1,	4.7.1	
Dielectric withstanding voltage	3.16	4.7.10	
Subgroup 4			
Solderability	3.10	4.7.4	See 4.6.1.2.5

1/ At the manufacturer's option, the determination of resistance change may be by any method which is within the accuracy requirements of this specification.
2/ Resistors shall meet the specified initial resistance tolerance after being subjected to the preceding test.

4.6.1.2.1 Subgroup 1. Subgroup 1 tests shall be performed on 100 percent of the product supplied under this specification. Resistors that are out of resistance tolerance, or which experience a change in resistance greater than that permitted for the tests of this subgroup shall be removed from the lot. Only lots having not more than 5 percent rejects or one resistor, whichever is greater due to exceeding the specified resistance change limit, as a result of subgroup 1 tests shall be furnished on orders. Corrective action shall be taken on such values and new pieces furnished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.6.1.2.2 Manufacturers production inspection. If the manufacturer performs tests equal to or more stringent than those specified in subgroup 1, table XII, as the final step of this production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than that specified for subgroup 1. Test conditions shall be equal to or more stringent than those specified for subgroup 1.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than that specified herein.

d. The lot rejection criteria is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. This data shall be provided as part of the evaluation required for MIL-STD-790. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.

f. Once approved the manufacturer shall not change the test procedures or criteria without prior notification and concurrence by the qualifying activity.

4.6.1.2.3 Subgroup 2 tests (PPM categories).

4.6.1.2.3.1 Sampling plan.

4.6.1.2.3.1.1 PPM-2. PPM-2 dc resistance test shall be performed on an inspection lot basis. Samples subjected to subgroup 2 PPM-2 shall be selected in accordance with table XIII based on the size of the inspection lot. (NOTE: larger samples may be inspected by the contractor in order to calculate PPM, however rejection of the lot shall be based on 1 or more defects). In the event of one or more failures the lot shall be rejected. Equipment used to perform subgroup 2 tests shall not be the same as those used in the subgroup 1 100-percent tests.

4.6.1.2.3.1.2 PPM-3. PPM-3 visual and mechanical examination shall be performed on an inspection lot basis. A sample of 13 parts shall be randomly selected, if one or more defects are found the lot shall be rejected.

4.6.1.2.3.2 Rejected lots. The rejected lot shall be segregated from new lots and those lots that have passed inspection. The rejected lot shall be 100 percent inspected for those quality characteristics found defective in the sample and any defects found removed from the lot. A new sample of parts shall be randomly selected using the original sample size (see 4.6.1.2.3.1). If one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.3.3 PPM calculations. PPM calculations shall be based on the results of the first sample check as prescribed in 4.6.1.2.3.1. Calculations and data exclusion shall be in accordance with EIA-554. (NOTE: PPM calculations shall not use data on the second sample submission.)

4.6.1.2.4 Subgroup 3. Subgroup 3 shall be performed on an inspection lot basis. A sample of 13 parts shall be randomly selected, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall be randomly selected, if one or more defects are found in this second sample, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.5 Subgroup 4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.6.1.2.5.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. The manufacturer may use electrical rejects from the subgroup 1 screening tests for all or part of the samples to be used for solderability testing. If there are one or more defects, the lot shall be considered to have failed.

4.6.1.2.5.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.7.4. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in (b).

b. The manufacturer submits the failed lot to a 100 percent solder dip using an approved solder dip process per 3.5.1.6. Following the solder dip the electrical measurements required in group A subgroup 1 tests shall be repeated on 100 percent of the lot. The Percent Defective Allowable (PDA) for the electrical measurements shall be as for the subgroup 1 tests. Thirteen additional samples shall then be selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

4.6.1.2.5.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract.

TABLE XIII. Sampling plans for PPM categories.

Lot size	Sample size for PPM -2/
0 - 30	100 percent
33 - 125	100 percent
126 - 3,200	125
3,201 - 10,000	200
10,001 - 35,000	315

4.6.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table XIV, in the order shown, and the sample shall be selected from inspection lots that have been subjected to and passed group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.6.1.3.1 Sampling plan. A sample of 13 parts shall then be randomly selected, if one or more defects are found, the lot shall be rescreened and defects removed. A new sample of 13 parts shall then be randomly selected, if one or more defects are found in the second sample, the lot shall be rejected and shall not be supplied to this specification.

TABLE XIV. Group B inspection.

Inspection	Requirement	Method	Number of samples
Resistance to solvents	3.11	4.7.5	13
Thermal shock	3.12	4.7.6	
Short time overload	3.15	4.7.9	
Insulation resistance	3.17	4.7.11	

4.6.1.3.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.6.2 Periodic inspection. Periodic inspection shall consist of group C inspection (see table XV). Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.3), delivery of products which have passed groups A and B shall not be delayed pending the results of these periodic inspection. A separate sample shall be selected from lots as defined in 4.6.1.1 for each enclosure.

4.6.2.1 Sampling plan.

4.6.2.1.1 Monthly (subgroup 1). A minimum of five samples shall be selected from each inspection lot and placed on extended life test of 4.7.16 once a month for the full 10,000-hour life test. A sufficient number of samples shall be selected from each lot by the manufacturer so that the maintenance of failure rate requirements are complied with; for each style separately for failure rates M and P, and for combined styles for failure rates R and S, within specified maintenance period. As far as is practicable, the manufacturer shall select the resistance values so that the full range of all resistance decades produced during the maintenance period are represented. The accumulated data shall be used for maintenance or extension of failure rate qualification (see 3.22.2).

4.6.2.1.1.2 Monthly subgroup 2. Each month the specified number of sample units shall be subjected to the inspections of table XV. The samples shall be selected from a lot as defined in 4.6.1.1, and where possible shall be representative of the styles included in the lot. Over a 6-month period, the manufacturer should select samples so that a maximum variety of styles produced are tested. A separate set of samples shall be tested for each enclosure material each month.

4.6.2.1.1.3 Semiannual. Sample units and tests shall be as specified in table XV. The test sample should include all styles allowed to be combined for lot formation (see 4.6.1.1) as practicable. A complete separate sample shall be selected for each enclosure.

4.6.2.1.1.4 Annual. Sample units and tests shall be as specified in table XV. The test sample shall include all styles allowed to be combined for lot formation (see 4.6.1.1) as practicable. A complete separate sample shall be selected for each enclosure material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.6.2.1.2 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or acquisition document.

TABLE XV. Group C inspection.

Inspection	Requirement paragraph	Method paragraph	Number of samples	Number of defects allowed
Monthly				
Subgroup I				
Life	3.22	4.7.16	See 4.6.2.1.1.1	See 4.6.2.1.1.1
Subgroup II				
Thermal shock	3.12	4.7.6	10 (6 highest, 4 lowest)	1
Resistance temperature characteristic	3.13	4.7.7		
Low temperature storage	3.14	4.7.8		
Dielectric withstanding voltage	3.16	4.7.10		
Insulation resistance	3.17	4.7.11		
Moisture resistance	3.18	4.7.12		
Terminal strength	3.19	4.7.13		
Semiannually				
Subgroup I				
Thermal shock	3.12	4.7.6	30 (15 highest, 15 lowest)	1
Shock, (specified pulse)	3.20	4.7.14		
Vibration, high frequency	3.21	4.7.15		
Subgroup II				
Reactance	3.25	4.7.19	30 (10 highest, 10-301 ohms, 10 lowest)	0
Annually				
High temperature exposure	3.23	4.7.17	102 (51 highest, 51 lowest)	1

4.6.2.1.3 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall immediately notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units or product which can be corrected and which were manufactured under essentially the same conditions with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action shall be furnished to the cognizant inspection activity and the qualifying activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.6.3 Inspection of packaging. The sample and inspection of preservation, packaging, and marking shall be in accordance with the requirements of MIL-R-39032.

4.7 Methods of inspections.

4.7.1 Visual and mechanical inspections. Resistors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4 to 3.5.1.7 inclusive, 3.26 to 3.28 and 3.28.1 inclusive).

4.7.2 Conditioning (see 3.8). Resistors shall be conditioned in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: Supported by their terminal leads at a point .375 ±.062 from the resistor body. Resistors shall be so arranged that the temperature of any one resistor shall not appreciably influence the temperature of any other resistor. There shall be no undue draft on the resistors. If forced-air circulation is employed, the air velocity shall not exceed 500 feet per minute and there shall be no direct impingment of the forced-air supply upon the resistors.

b. Temperature and tolerance: +25°C +15°C, -0°C.

c. Initial measurements: Initial resistance shall be measured as specified in 4.7.3 at +25°C +15°C, -0°C. This initial measurement shall be used as the reference for all subsequent measurements.

d. Operating conditions: Rated dc continuous working voltage or rated working voltage from an ac power supply at commercial line frequency, intermittently, 1 hour 30 minutes on and 30 minutes hour off for 100 +16, -4 hours. Where the resulting wave form is other than that of a commercial line, voltages shall be set using a "true rms" voltmeter and the peak voltage shall not exceed 2.5 times the rated continuous working voltage. Each resistor shall dissipate a wattage equal to the wattage of the resistor.

e. Measurement after condition: Resistance shall be measured at the end of 100 +16, -4 hours as specified in 4.7.2c and 4.7.3. The same procedure for measurement shall be used for initial and final measurements. Leads may be cleaned prior to final measurement.

f. Examination after conditioning: Resistors shall be examined for evidence of mechanical damage.

4.7.3 DC resistance (see 3.9). Resistors shall be tested in accordance with method 303 of MIL-STD-202. The following details and exceptions shall apply:

a. Measuring apparatus: Bridges or multimeters.

b. Limit of error of measuring apparatus: One fourth of the specified initial resistance tolerance or 0.1 percent, whichever is less, +0.002 ohm. The same measuring apparatus shall be used for any one test, but not necessarily for all tests.

c. Test voltage: Measurements of resistance shall be made by using a dc potential resulting in not more than 1-percent of rated wattage. This same voltage shall be used whenever a subsequent resistance measurement is made. For values less than 10 ohms, use a measuring device limited to 100 milliamperes or less.

d. Points of application of test voltage for initial resistance tolerance measurement: Resistors shall be measured by attaching the test leads as close to the resistor body as practical. For resistors 20 ohms and less, the referee point for attaching the test leads shall be .375 ±.062 inch from the body.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.7.4 Solderability (applicable to S and N terminals only) (see 3.10). Resistors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

a. Both leads shall be tested.

b. The leads shall be dipped within .062 inch of the body.

4.7.5 Resistance to solvents (see 3.11). Resistors shall be tested in accordance with method 215 of MIL-STD-202. The following details and exceptions shall apply:

a. The marked portion of the resistor body shall be brushed.

b. The number of sample units shall be as specified in tables XI and XIV, as applicable.

c. Resistors shall be examined for mechanical damage and legibility of markings.

4.7.6 Thermal shock (see 3.12). Resistors shall tested in accordance with method 107 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: In such a manner that there is at least 1 inch of free air space around each resistor, and in such a position with respect to the air stream that the mounting offers substantially no obstruction to the flow of air across and around the resistors.

b. Measurement before cycling: DC resistance shall be measured as specified in 4.7.3.

c. Test condition B.

d. Measurement after cycling: Not less than 1 hour, but within a 24-hour period after the last cycle, dc resistance shall be measured as specified in 4.7.3.

e. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.7 Resistance-temperature characteristic (see 3.13). Resistors shall be tested in accordance with method 304 of MIL-STD-202. The test temperature shall be in accordance with table XVI.

TABLE XVI. Temperature for resistance-temperature characteristic test.

Sequence	Temperature °C	
	For qualification inspection	For quality conformance inspection
1	25 1/	25 1/
2	-15	---
3	-55	-55
4	25 1/	25 1/
5	125	125
6	200	---
7	275	275

1/ This temperature shall be considered the referee temperature for each of the suceeding temperatures.

NOTE: At the option of the manufacturer the reverse sequence of table XVI may be as follows:

a. Room temperature.
b. +275°C.
c. Room temperature.
d. -55°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.7.8 Low-temperature storage (see 3.14).

4.7.8.1 Mounting. Resistors shall be mounted in such a manner that there is at least 1 inch of free air space around each resistor, and in such a position with respect to the air stream that the mounting offers substantially no obstruction to the air across and around the resistors.

4.7.8.2 Procedure. DC resistance shall be measured as specified in 4.7.3. Within 1 hour after this measurement, the resistors shall be placed in a cold chamber at a temperature of -65°C ±2°C for a period of 24 ±4 hours. The resistors shall then be removed from the chamber and maintained at a temperature of +25°C ±5°C for a period of approximately from 2 to 8 hours; the dc resistance shall again be measured as specified in 4.7.3. Resistors shall then be examined for evidence of mechanical damage.

4.7.9 Short-time overload (see 3.15). DC resistance shall be measured as specified in 4.7.3. The resistors shall then be mounted by means other than soldering and shall be subjected to an overload voltage which will result in the specified multiple of rated wattage for a 5-second duration (see 3.1). DC resistance shall again be measured after the resistors have cooled to room temperature.

4.7.10 Dielectric withstanding voltage (3.16).

4.7.10.1 Atmospheric pressure. Resistors shall be tested in accordance with method 301 of MIL-STD-202. The following details and exceptions shall apply:

a. Special preparations: Resistors shall be placed in a conductive material which will conform to the resistor surface so that between 90 and 100 percent of the outer periphery is contacted. The conductive material shall be centered on the resistor body. Care should be taken that any part of the resistor lead is as far away from the conductive material as possible.

b. Initial measurement: DC resistance shall be measured as specified in 4.7.3. Not applicable for group A inspection.

c. Magnitude of test voltage: As specified (see 3.1).

d. Nature of potential: An alternating current (ac) supply at commercial line frequency and waveform.

e. Duration of application of test voltage:

(1) Five seconds for group A inspection.

(2) One minute for qualification and group C inspection.

f. Rate of application of voltage: The test voltage shall be raised from zero to the application value, as uniformly as practicable, at the following rates:

(1) 500 volts rms per second for group A testing.

(2) 100 volts rms per second for group C and qualification testing.

g. Points of application of test voltage: Between the resistor terminals connected together and mounting hardware, or the conductive material, as applicable.

h. Measurement during the test: The leakage current shall be monitored during the application of test voltage.

i. Measurement after test (not applicable to group A testing): DC resistance shall be measured as specified in 4.7.3.

j. Examinations after test: Resistors shall be examined for evidence of flashover, mechanical damage, arcing, and insulation breakdown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.7.10.2 Barometric pressure (reduced). Resistors shall be tested in accordance with method 105 of MIL-STD-202. The following details and exceptions shall apply: (Applicable to qualification and group C inspection only.)

a. Method of mounting: As specified in 4.7.10.1a.

b. Initial measurement: DC resistance shall be measured as specified in 4.7.3.

c. Test condition D (100,000 feet).

d. Test voltage during subjection to reduced pressure: As specified (see 3.1).

e. Nature of potential: As specified in 4.7.10.1d.

f. Duration of test: One minute.

g. Points of application of test voltage: As specified in 4.7.10.1g.

h. Final measurement: DC resistance shall be measured as specified in 4.7.3.

i. Examinations after test: As specified in 4.7.10.1i.

4.7.11 Insulation resistance (see 3.17). Resistors shall be tested in accordance with method 302 of MIL-STD-202 with the following exceptions:

a. Test condition A.

b. Special preparation: In accordance with 4.7.10.1a.

c. Points of measurement: In accordance with 4.7.10.1g.

4.7.12 Moisture resistance (see 3.18). Resistors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: Soldered by the leads to rigid mounts or terminal lugs. The spacing of the mounts or terminal lugs shall be such that the length of each resistor lead is approximately .375 inch when measured from the edge of the supporting terminal to the resistor body. One half of the sample units shall be covered with a V shaped metal strap whose width is equal to the length of the resistor body as indicated on figure 1. The strap shall be made of a corrosion resistant metal and shall be kept in contact with the resistor body by supporting the body as indicted on figure 1, with a conducting, noncorrosive support whose width is less than that of the body and which will not act as a moisture trap. For group C inspection, each half of the sample shall be apportioned as three highest values, two lowest values for a total of five sample units per half. The mounting straps may be individual for each resistor or continuous for all resistors.

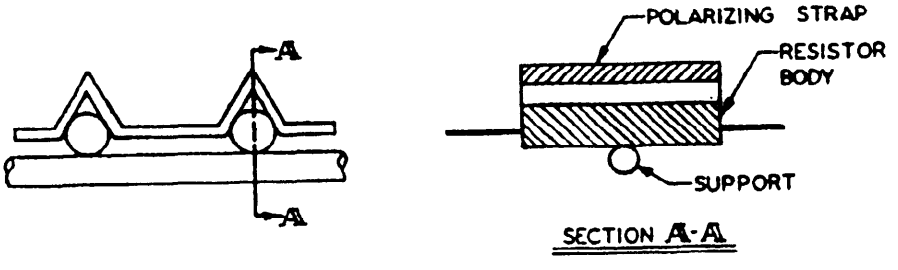


FIGURE 1. Mounting straps for moisture resistance test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

b. Initial measurement: Immediately following the initial drying period, dc resistance shall measured as specified in 4.7.3 at test conditions specified in 4.3.1.

c. Polarization and loading voltage:

(1) Polarization voltage: during steps 1 to 6 inclusive, a 100 volt dc potential shall applied only to those resistors which have a polarization strap. This potential shall be applied with the positive lead connected to the resistor terminals tied together, and the negative lead connected to the polarization straps.

(2) Loading voltage: During the first 2 hours of steps 1 and 4, a dc test potential equivalent to 100-percent rated wattage shall be applied to those resistors which do not have the polarizing strap specified in 4.7.12(a).

d. Subcycle: Step 7b shall not be applicable. Step 7a shall performed during any five of the first (9) cycles only.

e. Final measurements: Upon completion of step 6 of the final cycle, the resistors shall be held at the high humidity condition and a temperature of +25°C ±2°C for a period of 1 hour, 30 minutes to 3 hours, 30 minutes. Resistors shall then be removed from the chamber, and within 30 minutes, the dielectric withstanding voltage (atmospheric), insulation resistance, and dc resistance tests shall be performed as specified in 4.7.10.1, 4.7.11, and 4.7.3, respectively. The same straps used for polarizing the resistors may also be used for the dielectric withstanding voltage and insulation resistance tests.

f. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.13 Terminal strength (see 3.19). Tests shall be in accordance with method 211 of MIL-STD-202 with the following exceptions:

a. Test condition A and D. (Pull test and twist test, respectively.)

b. Measurement before test (A): DC resistance as specified in 4.7.3.

c. Method of holding (A): Resistors shall be clamped by one terminal lead.

d. Applied force (A): As specified (see 3.1).

e. Measurement after test (D): DC resistance as specified in 4.7.3 and examined for evidence of breaking and loosening of terminals and chipping of coating.

4.7.14 Shock, (specified pulse)(see 3.20). Resistors shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Special mounting means: Resistors shall be rigidly mounted on appropriate jig fixtures with their leads supported at a distance of .375 ±.062 inch from the resistor body. These fixtures shall be constructed in a manner to insure that the points of the resistor mounting supports will have the same motion as the shock table. The leads used during this test shall be no larger than AWG size 22 stranded wire, so that the influence of the test lead on the resistor will be held to a minimum. The test lead length shall be no longer than necessary. In all cases, the resistors shall be mounted in relation to the test equipment in such a manner that the stress applied is in the direction which would be considered most detrimental.

b. Measurement before shock: DC resistance shall be measured as specified in 4.7.3.

c. Test condition I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

d. Number and direction of applied shocks: The resistors shall be subjected to a total of 10 shocks in each of two mutually perpendicular planes, one perpendicular and the other parallel to the longitudinal axis of the resistor.

e. Measurement during shock: DC resistance shall be measured as specified in 4.7.3.

f. Measurement after shock: DC resistance shall be measured as specified in 4.7.3.

g. Examination after test: Resistors shall be examined for evidence of mechanical and electrical damage.

4.7.15 Vibration, high frequency (see 3.21). Resistors shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting specimens: Resistors shall be mounted on appropriate jig fixtures with their bodies restrained from movement and their leads supported at a distance of .375 ±.062 inch from the resistor body. The fixtures shall be constructed in a manner to insure that the points of the resistor mounting supports will have the same motion as the vibration test table. The fixtures shall also be of a construction that will preclude any resonance in the fixture when subjected to vibration within the test frequency range, and the fixture shall be monitored for these features on the vibration table. Test leads used during this test shall be no larger than AWG size 22 stranded wire, so that the influence of the test lead on the resistor will be held to a minimum. The test lead length shall be no greater than is absolutely necessary. A shielded cable, which may be necessary because of the field surrounding the vibration table, shall be clamped to the resistor mounting jig.

b. Initial measurement: DC resistance shall measured as specified in 4.7.3.

c. Test condition D.

d. Direction of motion: In each of two mutually perpendicular directions, one perpendicular and the other parallel to the longitudinal axis of the resistor. Total test time shall be 6 hours in each direction for a total of 12 hours.

e. Measurement during test: Each resistor shall monitored to determine electrical discontinuity by a method which shall at least be sensitive enough to monitor or register, automatically, any electrical discontinuity of 0.1 millisecond or greater duration.

f. Measurement after vibration: DC resistance shall be measured as specified in 4.7.3.

g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.16 Life (see 3.22). Resistors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

a. Method of mounting: Resistors shall be mounted on lightweight terminals. The integrity of the terminations shall be determined at each measurement interval. The voltage applied to any resistor shall not be less than 95 percent of the dc, ac line or true rms rate continuous working voltage. Resistors shall be arranged that the temperature of any one resistor shall not appreciably influence the temperature of any other resistor. If force air circulation is employed, the air velocity shall not exceed 500-feet per minute, and there shall be no direct impingement of the forced air supply on the resistors.

b. Test temperature: +25°C ±5°C.

c. Initial measurements: Initial resistance shall be measured after mounting at +25°C ±5°C. This initial measurement shall used as the reference temperature for all measurements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

d. Operating conditions: Resistors shall be operated at full rated wattage by applying dc continuous working voltage, or ac rated continuous working voltage from an ac supply at commercial line frequency, intermittently, 1 hour 30 minutes "on" and 30 minutes "off" for the applicable number of hours (see 4.7.16f). "On time" shall be .750 of the total elapsed time. The actual test time shall be recorded. Where the resulting waveform is other than that of a commercial line, voltages shall be set using a "true rms" voltmeter, and the peak voltage shall not exceed 2.5 times the rated continuous working voltage.

e. Test condition: 2,000 hour for qualification inspection with all samples continued on test to 10,000 hours. 10,000 hours for failure rate level determination of group C.

f. Measurements during test:

(1) Qualification inspection: Resistance (see 4.7.3) shall be measured at the end of the 30 minutes "off" periods after 250 +72, -24, 500 +72, -24, 1,000 +72, -24, and 2,000 +96, -24 hours have elapsed. Units continued on test shall be measured at intervals above 2,000 hours in accordance with 4.7.16f(2).

(2) Extended life testing: Resistance (see 4.7.3) shall be measured at the end of the 30 minutes "off" periods after 250 +72, -24, 500 +72, -24, 1,000 +72, -24, 2,000 +96, -24 and every 2,000 +96, -0 hours thereafter, until the required extended life period (10,000 hours) has elapsed. Measurements shall be made as near as possible to the specified time but may be adjusted so that measurements need not be made during other than normal working days.

g. Examination after test: Resistors shall be examined for evidence of mechanical damage.

4.7.17 High-temperature exposure (see 3.23).

a. Mounting: Resistors shall be mounted by their normal mounting means.

b. Initial measurements: DC resistance shall be measured as specified in 4.7.3 at room ambient conditions.

c. Procedure: Following initial resistance measurements, resistors shall be placed in a chamber maintained at +275°C ±7°C for a period of 2,000 hours +72, -0 hours with no load applied.

d. Measurements during test: At 250 +48,-0 hours, resistors shall be removed from the chamber and permitted to stabilize at room temperature and within 6 hours after removal, the dc resistance shall be measured a specified in 4.7.3. Resistors shall be examined for evidence of mechanical damage.

e. Final measurements: After removal from the test chamber, resistors shall be permitted to stabilize at room ambient temperature and within 6 hours after removal, cleaning of the leads will be allowed and the dc resistance shall be measured as specified in 4.7.3. Resistors shall be examined for evidence of mechanical damage.

4.7.18 Fungus (see 3.24). Unless certification is provided, resistors shall be tested in accordance with method 508 of MIL-STD-810.

4.7.19 Reactance (applicable to "N" and "Z" terminals and windings only) (see 3.25). Effective series inductance and parallel capacitance shall be determined using methods I and II, as applicable (see 4.7.19.1 and 4.7.19.2). The units shall be mounted by their normal mounting means using the fixtures as established in the following procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

4.7.19.1 Method I. This method shall be used for resistance values of 100 ohms and below. Use Boonton electronics model 63H or equivalent. Effective inductance of the resistors shall be measured at a frequency of 0.5 megahertz using the test fixtures TF-A or TF-B, as applicable and shown on figure 2. The appropriate test fixture shall be mounted to the binding post of the test bridge. The appropriate shorting bar conforming to figure 3 shall be inserted in the clips of the test fixtures so that the terminals rest firmly against the stops and the shorting bar is centered between the test fixture terminals. The bridge shall then be nulled and calibrated prior to the inductance readings. After removing the shorting bar, insert the resistor to be tested so that the leads are straight and rest firmly against the stops and the unit is centered between the test fixture terminals. The bridge shall be adjusted until the meter indicates a null. The inductance can be read directly from the bridge.

4.7.19.2 Method II. This method shall be used on resistance values above 100 ohms. Use Boonton radio RX meter model 250A or equivalent. Effective inductance of the resistors shall be measured at a frequency of 1 megahertz using the test fixtures TF-C or TF-D, as applicable, and shown on figure 4. The appropriate test fixture shall be mounted to the binding posts of the RX meter. The appropriate shorting bar conforming to figure 3 shall be inserted in the clips of the test fixtures so that the terminals rest firmly against the stops and the shorting bar is centered between the test fixture terminals. The bridge shall then be nulled and calibrated prior to inductance readings. After removing the shorting bar, insert the resistor to be tested so that the leads are straight and rest firmly against the stops and the unit is centered between the test fixture terminals. The bridge shall be adjusted until the meter indicated a null.

4.7.19.3 Cp readings. If the Cp reading falls within the:

a. Positive portion of the dial, the resistor is capacitive and the equivalent parallel capacitance is read directly from the dial.

b. Negative portion of the dial, the resistor is inductive and the equivalent series inductance can be computed using the following formulas:

$$Q = 2 \quad f \; Rp \; CP$$

$$Ls = \frac{Cp \; Rp^2}{1 + Q^2}$$

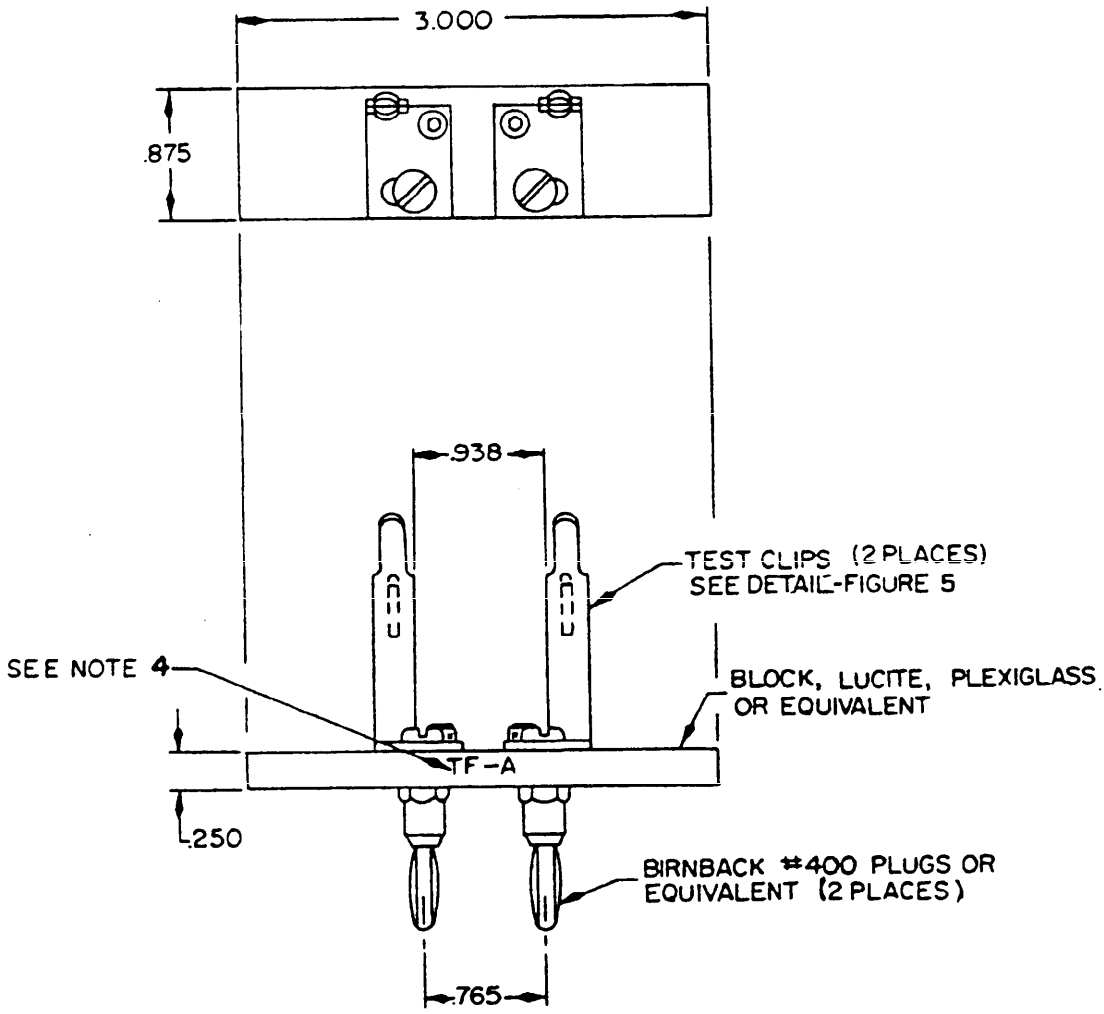
Where: f = frequency in hertz
Rp = indicated parallel resistance
Cp = indicated parallel capacitance
Ls = Equivalent series inductance

4.7.19.4 Examination after test. Units shall not exceed the limits specified in 3.25.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-39032.

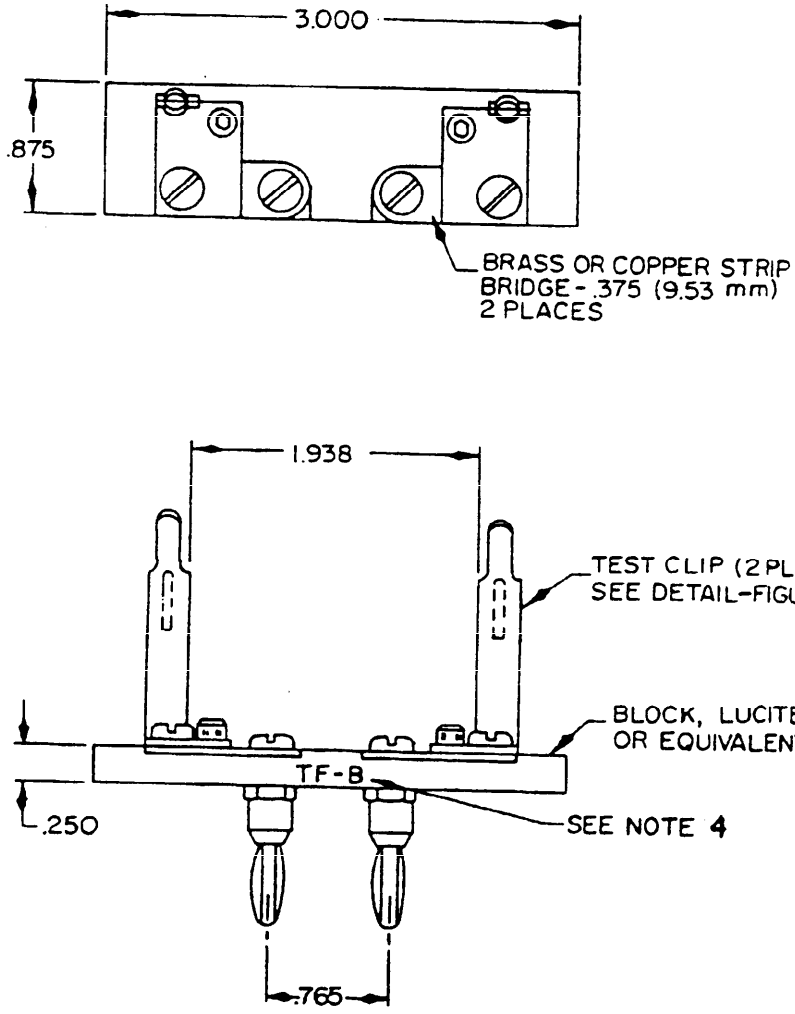
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.



TF-A fixture

FIGURE 2. Test fixture for method I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.



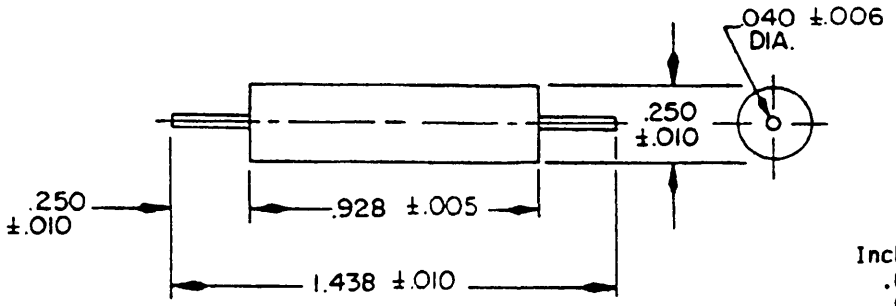
Inches	mm
.005	0.13
.250	6.35
.375	9.53
.765	19.43
.875	22.23
.900	22.86
1.938	49.23
3.000	76.20

TF-B fixture

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. Identification to be marked on fixture.
5. TF-A fixture shall be used when resistor body length is .900 (22.86 mm) or less.
6. TF-B fixture shall be used when resistor body length is greater than .900 (22.86 mm).

FIGURE 2. Test fixture for method I - Continued.

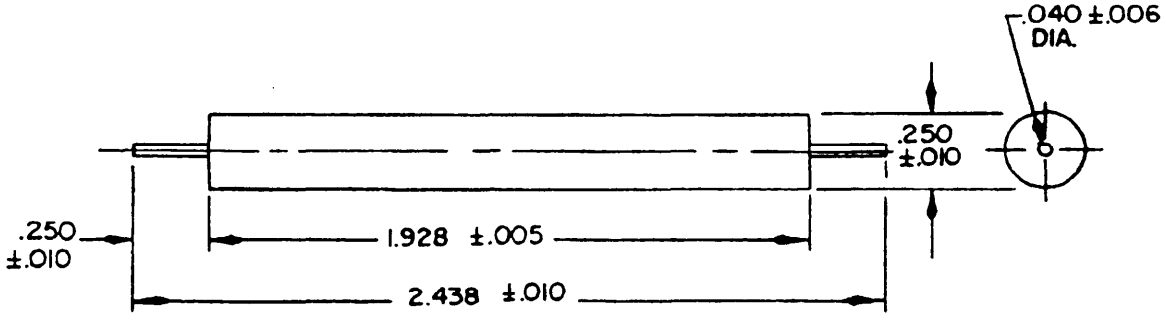
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.



COPPER OR BRASS

TF-A fixture and TF-C

Inches	mm
.005	0.13
.006	0.15
.010	0.25
.040	1.02
.250	6.35
.928	23.57
1.438	36.53
1.928	48.97
2.438	61.93



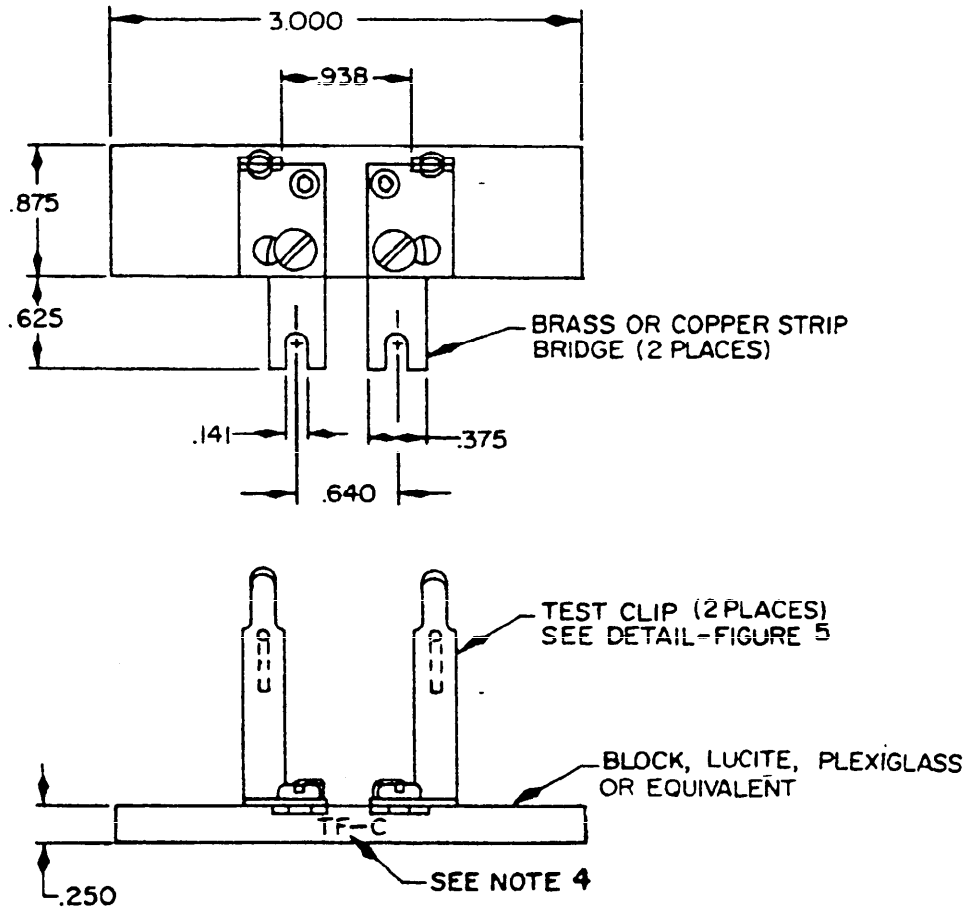
COPPER OR BRASS

TF-B fixture and TF-C

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Shorting bar for test fixtures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

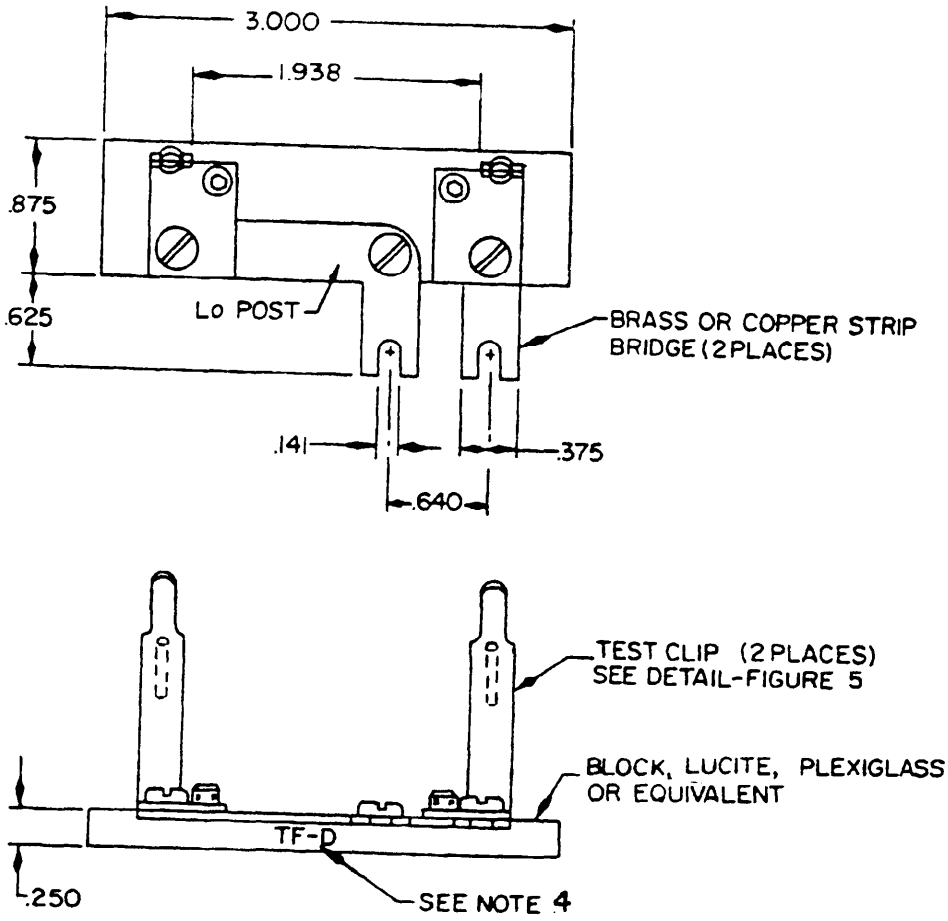


TF-C fixture

FIGURE 4. Test fixture for method II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

convenience of the reader.



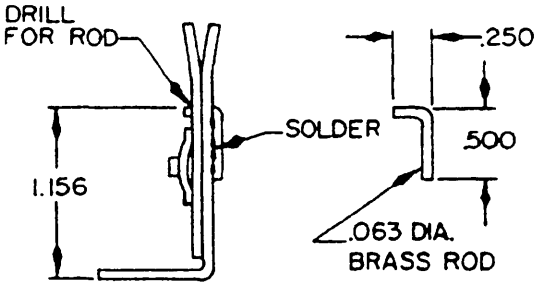
Inches	mm
.005	0.13
.141	3.58
.250	6.35
.375	9.53
.625	15.88
.640	16.26
.875	22.23
.900	22.86
.938	23.83
3.000	76.20

TF-D fixture

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).
4. Identification to be marked on fixture.
5. TF-C fixture shall be used when resistor body length is .900 (22.86 mm) or less.
6. TF-D fixture shall be used when resistor body length is greater than .900 (22.86 mm).

FIGURE 4. Test fixture for method II - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.



Inches	mm
.063	1.60
.250	6.35
.500	12.70
1.156	29.36

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 5. Test clip detail.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not manadatory.)

6.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 1.2.1 and 3.1).

c. Levels of preservation, packaging, and applicable marking (see section 5).

d. Marking requirement for failure rate level (see 3.27).

6.2 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the manufacturers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the uS Army Laboratory Command; however, information pertaining to qualification of products may be obtained from Defense Electronics Supply Center (DESC-E), 1507 Wilmington Pike, Dayton, OH 45444.

6.3 Derating. The intension of this specification is to cover resistors capable of full load operation with a high degree of stability at any ambient temperature up to and including +25°C. However, if it is desired to operate these resistors at ambient temperatures greater than +25°C, the resistors should be derated in accordance with figure 6.

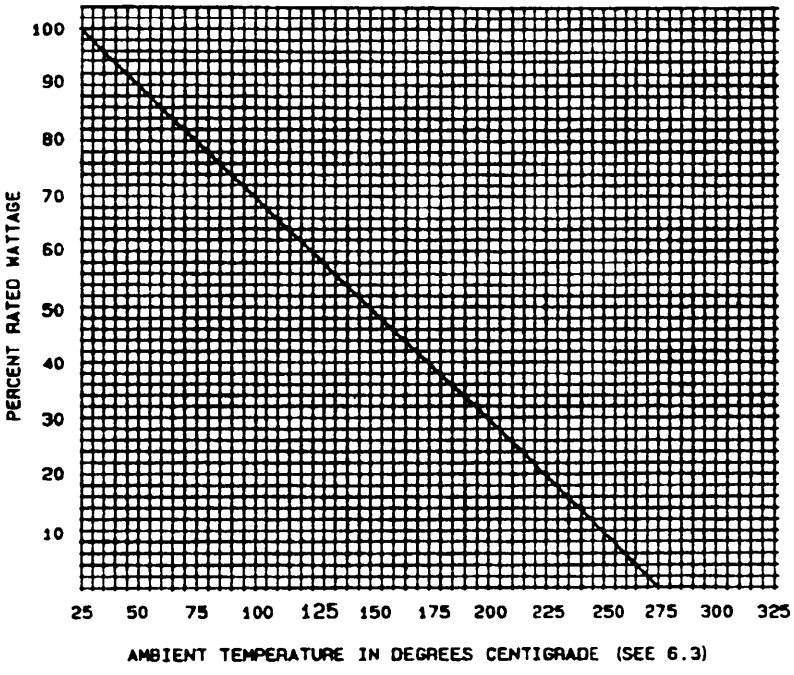


FIGURE 6. Derating curve for ambient temperatures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

6.4 Mounting. Under conditions of severe shock or vibration, or a combination of both, resistors of all sizes described in this specification should be mounted so that the body of the resistor is restrained from movement with respect to the mounting base. It should be noted that if clamps are used, certain electrical characteristics of the resistor will be altered. The heat dissipating qualities of the resistor will be enhanced or retarded depending on whether the clamping material is a good or poor heat conductor. Under less severe vibration conditions, all styles may be supported by their leads only. The lead lengths should be kept as short as possible, .250 inch or less preferred, but not longer than .625 inch. The longer the lead, the more likely that a mechanical failure will occur.

6.4.1 RWR80 series mounting. When RWR80 series resistors are to be used at full rated wattage, the maximum lead length shall be 0.500 inch from the resistor body to the mounting surface.

6.5 High frequency. Resistors should not be used in circuits where their ac performance is of critical importance in the operation of such circuits.

6.6 Coating materials. Certain coating materials used in fabricating resistors to this specification may be subject to "outgassing" of volatile material when operated at surface temperatures over +200°C. This phenomena should be taken into consideration for equipment design.

6.7 Selection and use information. Equipment designers should refer to MIL-STD-199, "Resistors, Selection and Use of," for a selection of standard resistor types and values for any new equipment design. All applications and use information concerning these resistors are also provided in MIL-STD-199.

6.8 Thermal conductivity and cost (RWR80 series). The RWR80 series of resistors are constructed with cores of higher thermal conducting material than for the RWR70 series; as a result, the RWR80 series of resistors are higher in power ratings and cost in respect to the equivalent RWR70 series of resistors.

6.9 Maintenance and replacement of lead types. For maintenance and replacement or other logistical purposes, solderable leads (S) shall be used to support weldable leads (W). Weldable leads should not be stocked.

6.10 Supersession data.

6.10.1 Characteristic designation. Resistors formerly identified with characteristic letter "G" under MIL-R-39007 are interchangeable and are superseded by resistors of the same style and resistance value under MIL-R-39007D.

6.10.2 Style supersession. The styles of the specification supersede other styles and types as given in table XVII.

6.10.3 Method of winding. Resistors classified as having an inductive method of winding (see 1.2.1.2 and 3.5) are equivalent to resistors previously identified with the same PIN and with a solderable terminal (symbol S).

6.11 Retinning leads. If retinning (hot solder dip) of the leads is required see 3.5.1.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

TABLE XVII. Style supersession.

Styles in MIL-R-39007D	Superseded styles in MIL-R-39007A	Superseded styles in 1/ MIL-R-39007	Superseded detail specification in MIL-R-38101(USAF)
RWR71	RWR71	RWR71	MIL-R-38101/2
RWR74	RWR74	RWR67	MIL-R-38101/16
RWR78	RWR78	RWR68	MIL-R-38101/18
RWR80	RWR80	RWR70	MIL-R-38101/14
RWR81	RWR81	---	MIL-R-38101/13
RWR84	RWR84	---	MIL-R-38101/17
RWR89	RWR89	RWR69	MIL-R-38101/15

1/ Solderable lead type only.

6.12 Reactance resistors. When low reactance resistors are required with values below 10 ohms, the S and W windings can be used since these values have low reactance and meet the requirements specified in table VIII.

6.13 Caution in handling RWR80 series resistors. The cores of RWR80 series resistors are usually made of beryllium (or beryllium oxide). Dust from the core material of these resistors is hazardous if inhaled or deposited in cuts in the skin. Do not machine, abraid, file, cut etc. without proper safety equipment and procedures.

6.14 Subject term (key word) listing.

Resistor, fixed
Resistor, power type
Resistor, wirewound

6.15 PIN. The PIN is a new term encompassing previous terms used in specifications such as part number, type designator, identification number (see 1.2.1).

6.16 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.

APPENDIX

PROCEDURE FOR QUALIFICATION

10. SCOPE

10.1 Scope. This appendix details the procedure for submission of samples, with related data, for qualification testing and approval of resistors covered by this specification. The procedure for extending qualification approval of the required sample to other resistors covered by this specification is also outlined herein. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

30. SUBMISSION.

30.1 Sample. A sample consisting of 133 highest values 1/ and 133 lowest values 1/ (down to 0.10 ohm), coated or enclosed resistors in each style, terminal type or in each style terminal type (see 40.2), both using a nominal diameter wire of 0.001 inch (.0009 absolute diameter), having a resistance tolerance of ±1 percent shall be submitted in accordance with 40. Where the use of 0.0008 inch wire is authorized for an individual style (see 3.1), the highest value sample submitted may be constructed using 0.0008 inch wire. At the option of the manufacturer, tolerances of ±.1 or ±.5 may be submitted in lieu of ±1 percent. In addition, 10 sample units each, of any resistance value shall be submitted and subjected to the tests of groups II and VIII, respectively, and 10 lowest values, 10 highest value and 10 units of 301 ohms of noninductive wound to test group IX. If the same coating and core materials are not used, a separate submission shall be furnished for each in each coating and core material for which qualification approval is sought. Four uncoated or unenclosed (2 at the highest value and 2 at the lowest value down to 0.10 ohms), shall also be submitted in each style 2/. If enclosures are used in lieu of coatings, four enclosures shall be furnished (see 40.1).

30.1.1 Additional sample for extension of terminal qualification. When terminal type "S" in a style is submitted, qualification for type "W" shall be granted with the added submission of 32 type "W" resistors (16 of the highest and 16 of the lowest resistance values) to groups I, Ia, V (2 additional unenclosed units, one of the highest and one of the lowest resistance value) and III of table XI. When terminal type "W" in a style is submitted, qualification for type "S" shall be granted with the added submission of 32 (16 of the highest and 16 of the lowest resistance values) samples of type "S" to groups I, Ia, V (2 additional unenclosed) and III of table XI. An additional 10 samples shall also be subjected to group II alone (see 40.1).

30.1.2 Additional samples for extension of tolerance qualification. When a sample of ±1 percent tolerance is submitted under 30.1, and where qualification to a lower resistance tolerance is desired (i.e., ±.5, .1 percent), 10 additional samples units each of the highest value and lowest or 10 ohms, whichever is higher, in the lowest resistance tolerance for which qualification is sought shall be submitted and subjected to the inspection of groups I, Ia, and III of table XI (see 40.1).

30.2 Test data. Each submission shall be accompanied by test data covering nondestructive tests listed in table XI which are performed on the submitted specimens. All test data shall be submitted in duplicate.

1/ One additional sample unit of each resistance value shall be submitted to permit substitution for the allowable defect in group IA inspection.

2/ The uncoated and unenclosed resistors shall be individually packaged to preclude damage in shipment.

Source: https://assist.dla.mil -- Downloaded: 202640-21T16:22Z
Check the source to verify that this is the current version before use.

30.3 Description of items. The manufacturers shall submit a detailed description of the resistors being submitted for test, including a description of the wire used for the resistance element, type of coating or enclosure, and material used for the terminals. All submitted samples shall be representative of the manufacturer's normal production. After qualification has been granted, no changes shall be made in materials, design, or construction without prior notification to the qualifying activity.

40. EXTENSION OF QUALIFICATION

40.1 Qualification extension. The extension of qualification or resistance values for each style shall range between the highest and lowest values qualified. The extension of qualification between failure rate levels shall be as follows:

Failure rate level	Will qualify failure rate level
S	S, R, P, M
R	R, P, M
P	P, M
M	M

The extension of qualification between tolerances shall be as follows:

Tolerance qualified	Will qualify
B	B, D, F
D	D, F
F (see note)	F (see note)

The extension between terminal types shall be as specified in 30.1.1.

NOTE: See 30.1.2 for extension when additional samples are submitted.

40.2 Qualification. Qualification of "N" terminal (solderable, noninductively wound) will qualify "Z" terminal (weldable, noninductively wound), "S" terminal (solderable, inductively wound), and "W" terminal (weldable, inductively wound) resistors.

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities
Army - AR, MI
Navy - AS, OS
Air Force - 17, 80, 85
DLA - ES

User activities
Navy - CG, MC
Air Force - 19

Preparing activity
Army - ER

Agent
DLA - ES

(Project 5905-1191)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:22Z
Check the source to verify that this is the current version before use.